

Mert Cemri

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EDUCATION

- **University of California, Berkeley** Berkeley, CA, USA
Ph.D. in Electrical Engineering and Computer Sciences *August, 2023 - May 2028*
Advisors: Prof. Ion Stoica, Prof. Kannan Ramchandran and Prof. Alex Dimakis
CGPA: 4.0/4.0
- **Bilkent University** Ankara, Turkey
B.S. in Electrical and Electronics Engineering *September, 2019 - June, 2023*
CGPA: 4.0/4.0
- **École Polytechnique Fédérale de Lausanne (EPFL)** Lausanne, Switzerland
Exchange Semester in Computer Science *February, 2022 - July, 2022*

RESEARCH INTERESTS

- Machine Learning, Natural Language Processing, Multi-Agent Systems, Distributed Processing, Test-Time Scaling

PUBLICATIONS

- **Conference Papers**
 - **M. Cemri**, M. Pan, S. Yang, L. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, and I. Stoica, ‘Why Do Multi-Agent LLM Systems Fail?’, in *Advances in Neural Information Processing Systems (NeurIPS) 2025 Dataset and Benchmark Track*, 2025.
 - H. Bai, Y. Zhou, **M. Cemri**, J. Pan, A. Suhr, S. Levine, A. Kumar, ‘DigiRL: Training In-The-Wild Device-Control Agents with Autonomous Reinforcement Learning’, *Advances in Neural Information Processing Systems (NeurIPS) 2024 Main Track*, 2024.
 - **M. Cemri**, V. Bordignon, M. Kayaalp, V. Shumovskaia, A.H. Sayed, ‘Asynchronous Social Learning’, *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023.
- **Workshop Papers**
 - **M. Cemri**, M. Pan, S. Yang, L. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, and I. Stoica, ‘Why Do Multi-Agent LLM Systems Fail?’, in *Proc. ICLR 2025 Workshop on Building Trust in Language Models and Applications*, 2025.
 - **M. Cemri**, N. Rajaraman, R. Tiwari, X. Liu, K. Keutzer, I. Stoica, K. Ramchandran, A. Beirami, and Z. Sun, ‘SPECS: Faster Test-Time Scaling through Speculative Drafts’, in *Proc. ICML 2025 Workshop on Efficient Systems for Foundation Models*, 2025.
 - **M. Cemri**, A. Jalal, K. Ramchandran, ‘Discrete Diffusion Posterior Sampling for Protein Design’, *ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling*, 2024.
- **Journal Papers**
 - V. Shumovskaia, M. Kayaalp, **M. Cemri**, A.H. Sayed, ‘Discovering Influencers in Opinion Formation over Social Graphs’, *IEEE Open Journal of Signal Processing*, 2023.
- **Preprints**
 - **M. Cemri**, T. Çukur, A. Koç, ‘Unsupervised Simplification of Legal Texts’, arXiv preprint arXiv:2209. 00557, 2022.

RESEARCH EXPERIENCE

- **University of California, Berkeley** Berkeley, CA
Berkeley AI Research (BAIR) / SKY Lab - Graduate Researcher *August 2023 - Present*
 - Working on efficient deep generative model inference algorithms. In particular, for large scale LLM serving and inference, I design novel speculative decoding and multi-agent strategies to decrease latency and improve performance and alignment of LLMs.
 - Worked on developing novel discrete diffusion and posterior sampling guiding algorithms. In particular, a current application of interest is designing unseen protein sequences using discrete diffusion posterior sampling.

- **Ecole Polytechnic Federale Lausanne (EPFL)** Lausanne, Switzerland
Adaptive Systems Lab (ASL) - Undergraduate Researcher February 2022 - June 2023
 - Worked under the supervision of Prof. Ali H. Sayed on distributed optimization and social learning.
 - Developed the theoretical model of social learning for the asynchronous case. Also conducted research on identifying the most influential agent over a social graph, and recovering the graph topology by analyzing interactions among users in social networks.
- **National Magnetic Resonance Research Center (UMRAM)** Ankara, Turkey
ICON Lab - Undergraduate Researcher March 2021 - July 2023
 - Worked on the applications of neural networks to NLP problems and on developing novel tools to analyze graphical data using graph neural networks under the supervision of Prof. Tolga Çukur and Prof. Aykut Koç.

WORK EXPERIENCE

- **Apple Inc.** Cupertino, CA
AI/ML Team - Machine Learning Intern May 2025 - August 2025
 - Worked on post training Apple Foundation Model (AFM) for tool-using capabilities.
- **Titra Technology** Ankara, Turkey
Autonomous Driving Unit - Project Engineer September 2022 - June 2023
 - Worked on reinforcement learning (RL) based autopilot systems for fixed-winged unmanned aerial vehicles.
- **The Scientific and Technological Research Council of Turkey (TÜBİTAK)** Ankara, Turkey
Advanced Technologies Research Institute (ATRI) - Intern June 2021 - August 2021
 - Worked on electronic warfare technologies. Implemented pseudo-random number generation algorithms on an FPGA board (with VHDL), produced Gaussian noise, and harvested the noise on a Jupyter notebook.
 - Built a neural network on a PYNQ FPGA, and studied how deep learning algorithms are accelerated with FPGAs.

SKILLS

- **Languages:** English (Fluent, TOEFL IBT: 110/120), German (Intermediate, Goethe Zertifikat B1), Turkish (Native)
- **Programming Languages:** Python, Matlab, C/C++, VHDL, Assembly Language, L^AT_EX
- **Frameworks:** vLLM, verl, PyTorch, NumPy, Pandas, Keras, Linux

HONORS AND AWARDS

- EECS Departmental Fellowship, UC Berkeley 2023-2024
- Research Excellence Award, Bilkent University 2023
- Academic Excellence Award, Bilkent University 2023
- Merit scholarship due to outstanding academic success, Bilkent University 2021-2023
- Tuition scholarship due to success in University Entrance Exam, Bilkent University 2019-2023
- Bronze Medal in the International Baltic Sea Philosophical Essay Event 2018
- Silver Medal in National Science and Technology Projects Competition for High School Students 2018

PROJECTS

- **Reinforcement Learning based Autopilot** In this work, we develop an RL-based controller for the attitude and landing controls of a fixed wing unmanned air vehicle. We are using PPO to do online learning.
- **Meta Learning with Zeroth Order Oracle** Studied the zeroth-order (ZO) methods in a learning to learn (L2L) framework with respect to their generalizability. In particular, Mert analyzed the zero-shot performance of these methods on new datasets and applied several optimization tricks to improve the generalization of this framework. We showed that the enhancements we made accelerate the convergence of the algorithm, and help prevent overfitting.
- **Autonomous Vehicle (EPFL Human-Robot Tandem Race '22)** Developed an autonomous vehicle that can track a particular human, and participated in EPFL Human-Robot Tandem Race with this vehicle. For detection of a selected person, we used a combination of Yolov5 and Pifpaf models, and for tracking, we used DeepSort and ReID methods.
- **Transmitter-Receiver for a AWGN Chanel that Adds Random Rotation to Messages** Built a transmitter and a receiver that creates 6-bit latent representations of 7-bit ASCII characters, send it through a WGN channel that also adds a random phase (rotation) to the signals, and successfully recover original word of 7-bit ASCII characters.

- **Gaussian Noise using PYNQ-Z1 board** Implemented Taus-88 algorithm for pseudo-random number generation (PRNG) purposes on PYNQ-Z1 FPGA and generated a Gaussian noise by combining such PRNG algorithms using the Central Limit Theorem. Harvested this noise by building a DMA channel and demonstrated it on a Jupyter notebook.
- **Catch the Mole** Developed a 2-D game using VHDL as the hardware description language and a VGA screen to display the game.
- **TRC-10 transceiver** Developed a TRC-10 transceiver with a wavelength of 10 meters.
- **Breakout Game** Participated in an intense two-week training organized by Prof. Nick McKeown of Stanford University CS Department. In this training, He developed a 2-D motion game using Java.